Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 8

Title: Building Marketing Campaign Performance Dashboard using Power BI

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Building Marketing Campaign Performance Dashboard using Power BI

Problem Statement

Nowadays, organizations run multiple marketing campaigns across various channels and platforms, generating a large amount of campaign performance data. Analyzing this data manually is difficult and time-consuming. In this experiment, a marketing campaign dataset was used to create an interactive dashboard in Microsoft Power BI. The objective was to analyze campaign performance, clicks, impressions, conversion rate, and return on investment (ROI) across different campaign types, channels, locations, and customer segments so that business information could be understood more easily through visual reports.

Methodology / Steps Followed

Step 1: Data Collection

The dataset contained information such as

- Campaign_ID, Company, Campaign_Type
- Target_Audience, Duration
- Channel_Used, Location, Language
- Customer_Segment, Date
- Acquisition_Cost, Clicks, Impressions
- Engagement_Score, Conversion_Rate, ROI

Step 2: Data Import

- The dataset was imported into Power BI Desktop using the Get Data option. After loading the data, all the columns were checked to ensure they were imported correctly.

Step 3: Data Cleaning and Transformation

- Checked the dataset for missing values.
- Removed duplicate records wherever necessary.
- Verified and corrected data types such as Date, Text, and Decimal Number.
- Renamed columns for better readability.

Created calculated columns:

- Conversion Rate %
- Highest ROI
- Lowest ROI

Step 4: Data Modeling

- Total Campaigns

- Total Clicks
- Total Impressions
- Total Acquisition Cost
- Average Conversion Rate
- Average ROI

Step 5: Data Modeling

- Created KPI Cards for Total Campaigns, Total Clicks, Total Impressions, Average Conversion Rate, and Average ROI.
- Created Bar Chart for Average ROI by Campaign Type.
- Created Bar Chart for Total Campaigns by Location.
- Created Donut Chart for Total Campaigns by Customer Segment.
- Created Bar Chart for Average ROI by Company.
- Created Gauge Chart for Average, Minimum, and Maximum of ROI.
- Created Line Chart for Sum of Clicks and Sum of Impressions by Month.
- Added slicers for Campaign_Type, Channel_Used, Customer_Segment, and Date.
- Designed an interactive marketing campaign analysis dashboard.

Step 6: DAX Measures and Calculated Columns

The following DAX measures and calculated columns were created to analyze the marketing campaign data and generate key business insights.

An interactive Power BI dashboard was designed using different visualization techniques.

Measures used are:.

```
Total Campaigns = COUNT(marketing_campaign_dataset[Campaign_ID])
Total Clicks = SUM(marketing_campaign_dataset[Clicks])
Total Impressions = SUM(marketing_campaign_dataset[Impressions])
Average Conversion Rate =
AVERAGE(marketing_campaign_dataset[Conversion_Rate])
Average ROI = AVERAGE(marketing_campaign_dataset[ROI])
```

Creating a new column “Conversion Rate %”:

```
Conversion Rate % =
marketing_campaign_dataset[Conversion_Rate] * 100
```

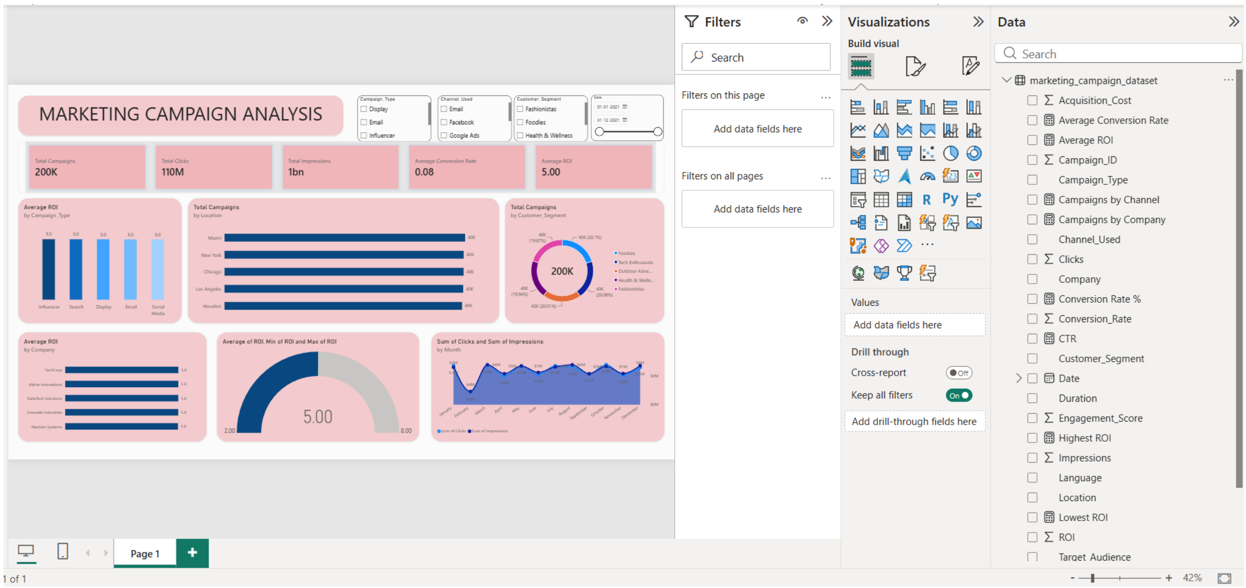
Creating new measures “Highest ROI” and “Lowest ROI”:

```
Highest ROI = MAX(marketing_campaign_dataset[ROI])
Lowest ROI = MIN(marketing_campaign_dataset[ROI])
```

Step 7: Dashboard Development

An interactive Power BI dashboard was designed using different visualization techniques.

Output



The final Power BI dashboard provides:

- Total Campaign Performance
- Channel-wise Campaign Analysis
- Location-wise Campaign Performance
- Customer Segment Analysis
- Company-wise ROI Comparison
- Monthly Clicks and Impressions Trend
- Interactive Filters and Slicers

The dashboard enables management to monitor marketing campaign performance and make data-driven decisions effectively.

Conclusion

This experiment successfully demonstrated the use of Microsoft Power BI for marketing campaign data analysis and visualization. The marketing campaign dataset was imported, cleaned, transformed, and analyzed using DAX measures and calculated columns. Various charts, KPI cards, and slicers were used to build an interactive dashboard that presents business information in a simple and meaningful way.

The dashboard helps users monitor campaign performance, compare ROI across companies and campaign types, analyze customer segments and locations, and observe monthly trends in clicks and impressions. Overall, the experiment improved the understanding of data visualization techniques and business intelligence concepts using Power BI.

Learning Outcomes

Through this experiment, I learned:

- How to import marketing campaign datasets into Microsoft Power BI.
- How to clean and transform business data using Power Query.
- How to verify and modify data types.
- How to create calculated columns.
- How to develop DAX measures for business analysis.
- How to build KPI cards for important business metrics.
- How to create Bar Charts, Donut Charts, Gauge Charts, and Line Charts.
- How to use slicers for interactive dashboard filtering.

https://github.com/MaharshiGoswami/Data-Analytics-for-business/tree/main/EXP9